Mod I: Basic Physiology MCQs Set—-
Some of explanations and answer key at the end

Part I:

1. **Which organelle is most abundantly found in the core of a cell's cilium?**
   a) Microfilaments
   b) Microtubules
   c) Mitochondria
   d) Ribosomes
   e) Lysosomes
2. **Maintenance of nearly constant conditions in the internal environment is defined as:**
   a) Osmosis
   b) Metabolism
   c) Homeostasis
   d) Phagocytosis
   e) Facilitated diffusion
3. **Programmed cell death is known as:**
   a) Necrosis
   b) Atrophy
   c) Hypertrophy
   d) Apoptosis
   e) Autophagy
4. **Which ion is most abundant in the extracellular fluid (ECF)?**
   a) Potassium
   b) Magnesium
   c) Phosphate
   d) Calcium
   e) Sodium
5. **The primary lipid constituent of the cell membrane bilayer is:**
   a) Triglyceride
   b) Phospholipid
   c) Cholesterol
   d) Glycolipid
   e) Sphingolipid
6. **The "Internal Environment" (Milieu Intérieur) refers specifically to:**
   a) Extracellular fluid
   b) Cytoplasm
   c) Intracellular fluid
   d) Nucleoplasm
   e) Blood plasma only
7. **The movement of water across a semipermeable membrane due to a concentration gradient is:**
   a) Exocytosis
   b) Pinocytosis
   c) Osmosis
   d) Active transport
   e) Simple diffusion
8. **The glycocalyx of the cell membrane is primarily composed of:**
   a) Pure lipids
   b) Pure proteins
   c) RNA

d) DNA
e) Carbohydrates

9. **The Sodium-Potassium (Na^+-K^+) ATPase pump moves how many ions?**
   a) 2 Na+ out, 3 K+ in
   b) 1 Na+ out, 1 K+ in
   c) 2 Na+ in, 2 K+ out
   d) 3 Na+ out, 2 K+ in
   e) 3 Na+ in, 2 K+ out
10. **Which mechanism is a "vicious cycle" that can lead to death if not controlled?**
   a) Adaptive control
   b) Negative feedback
   c) Feed-forward control
   d) Positive feedback
   e) Osmotic equilibrium
11. **Which transport process requires a carrier protein but NO energy?**
   a) Primary active transport
   b) Secondary active transport
   c) Simple diffusion
   d) Endocytosis
   e) Facilitated diffusion
12. **Which organelle is known as the "powerhouse" of the cell due to ATP production?**
   a) Nucleus
   b) Lysosome
   c) Golgi apparatus
   d) Endoplasmic reticulum
   e) Mitochondria
13. **Which of the following describes the percentage of body weight comprised by total body water in an average 70 kg man?**
   a) 40%
   b) 50%
   c) 60%
   d) 70%
   e) 80%
14. **The principal buffer in the Extracellular Fluid is:**
   a) Phosphate buffer
   b) Protein buffer
   c) Bicarbonate buffer
   d) Hemoglobin buffer
   e) Sulfate buffer
15. **What is the normal resting membrane potential of a large nerve fiber?**
   a) -90 mV
   b) -70 mV
   c) -55 mV
   d) +35 mV
   e) 0 mV
16. **Which of the following is an example of an "Adaptive Control" mechanism?**
   a) Baroreceptor reflex
   b) Blood clotting
   c) Childbirth
   d) Delay-rectified movement (Cerebellum)
   e) Insulin secretion
17. **Which organelle contains digestive enzymes for intracellular debris?**
   a) Nucleolus
   b) Peroxisome

c) Centriole
d) Ribosome
e) Lysosome

18. **The largest fluid compartment in the human body is:**
    a) Plasma
    b) Extracellular fluid
    c) Intracellular fluid
    d) Interstitial fluid
    e) Transcellular fluid
19. **Rough Endoplasmic Reticulum is "rough" because of the presence of:**
    a) Vesicles
    b) Golgi bodies
    c) Lipids
    d) Ribosomes
    e) Mitochondria
20. **Which transport mechanism moves glucose against its concentration gradient in the renal tubules?**
    a) Simple diffusion
    b) Facilitated diffusion
    c) Primary active transport
    d) Secondary active transport (Co-transport)
    e) Osmosis

Part II:

21. **If a cell with an internal osmolarity of 282 mOsm/L is placed in a solution of 350 mOsm/L NaCl, the cell will:**
    a) Swell and burst
    b) Stay the same size
    c) Actively pump Na+ in
    d) Shrink due to osmosis
    e) Undergo hemolysis
22. **In the Na^+-Ca^{2+} counter-transport mechanism, the energy to move Calcium out of the cell comes from:**
    a) Direct ATP hydrolysis
    b) GTP hydrolysis
    c) Thermal energy
    d) The concentration gradient of Na^+
    e) The concentration gradient of K^+
23. **The "Gain" of a control system is calculated as:**
    a) Error / Correction
    b) Input + Output
    c) Correction / Error
    d) (Error $\times$ Correction) / 2
    e) Stimulus / Response
24. **Which of the following would significantly INCREASE the rate of net diffusion of a substance across a membrane?**
    a) Increasing membrane thickness
    b) Decreasing the concentration gradient
    c) Decreasing the surface area of the membrane
    d) Increasing the lipid solubility of the substance
    e) Decreasing the temperature

25. **The "resting" cell membrane is 50 to 100 times more permeable to which ion than the other?**
    a) Sodium than Potassium
    b) Calcium than Sodium
    c) Potassium than Sodium
    d) Chloride than Potassium
    e) Magnesium than Calcium
26. **What would happen to the resting membrane potential if the extracellular Potassium concentration was doubled?**
    a) It would become more negative (hyperpolarize)
    b) It would not change
    c) It would reach +61 mV
    d) It would become less negative (depolarize)
    e) The cell would immediately die
27. **Digitalis (a drug) inhibits the Na^+-K^+ ATPase pump. This would directly result in:**
    a) Increased intracellular Sodium
    b) Decreased intracellular Sodium
    c) Increased intracellular Potassium
    d) Hyperpolarization of the cell
    e) Increased rate of glucose co-transport
28. **During the "Upstroke" of an action potential, the membrane potential moves toward:**
    a) The Nernst potential for Potassium
    b) 0 mV and stops
    c) The Nernst potential for Sodium
    d) -90 mV
    e) The Nernst potential for Calcium
29. **Which of the following is true regarding the "Refractory Period"?**
    a) A second AP can occur during the absolute refractory period
    b) The relative refractory period occurs during Phase 0
    c) It allows for tetanization of cardiac muscle
    d) It is caused by the inactivation of Sodium channels
    e) It only occurs in unmyelinated fibers
30. **If the "Gain" of the baroreceptor system is -2, and a sudden stimulus increases BP by 30 mmHg, the system will correct it by:**
    a) 10 mmHg
    b) 30 mmHg
    c) 60 mmHg
    d) 20 mmHg
    e) 15 mmHg
31. **The selective permeability of the K^+ "Leak" channel is due to:**
    a) A broad hydrated diameter
    b) A negative charge at the opening
    c) The presence of a voltage gate
    d) Carbonyl oxygens in the selectivity filter
    e) Simple lipid solubility
32. **A "Ligand-gated" channel opens in response to:**
    a) Change in voltage
    b) Mechanical stretch
    c) Binding of a chemical substance (e.g., ACh)
    d) Change in temperature
    e) Osmotic pressure
33. **In the process of Pinocytosis, the "clathrin" proteins are located on:**
    a) The extracellular surface of the membrane

b) The Golgi apparatus
c) The nuclear membrane
d) The inside of the cell membrane (cytoplasmic side)
e) The surface of mitochondria

34. **Total Body Water calculation: A 70 kg man has an ICF of 28 Liters. His ECF volume is approximately:**
    a) 5 Liters
    b) 10 Liters
    c) 25 Liters
    d) 14 Liters
    e) 42 Liters
35. **Which transport mechanism is saturated (reaches V_{max}) as the concentration of the substance increases?**
    a) Simple diffusion through a channel
    b) Simple diffusion through the lipid bilayer
    c) Osmosis
    d) Facilitated diffusion
    e) All of the above
36. **The "Fluid Mosaic Model" implies that:**
    a) Proteins are fixed in a solid lipid sheet
    b) Lipids are fixed and proteins move
    c) Both lipids and proteins can diffuse laterally
    d) The membrane is a rigid wall
    e) Water flows through the center of the bilayer
37. **A patient with a defect in the "SGLT" transporter would have difficulty with:**
    a) Pumping Sodium out of the cell
    b) Reabsorbing Glucose in the kidneys
    c) Contracting skeletal muscle
    d) Producing ATP
    e) Secreting Insulin
38. **The "Goldman Equation" is used to calculate:**
    a) The rate of diffusion
    b) The osmotic pressure
    c) The GFR
    d) The membrane potential when multiple ions are involved
    e) The cardiac output
39. **Which of the following inhibits the release of Cytochrome C from mitochondria to prevent Apoptosis?**
    a) Caspases
    b) Clathrin
    c) Actin
    d) Bcl-2 family (anti-apoptotic members)
    e) Hydrolases
40. **The thickness of the typical cell membrane is approximately:**
    a) 7.5 to 10 nanometers
    b) 7.5 to 10 micrometers
    c) 1 to 2 millimeters
    d) 100 nanometers
    e) 0.5 nanometers

Part III: Clinical Scenario Questions

41. **A patient arrives with severe dehydration. To expand their ECF volume without changing the ICF volume, you should administer:**
    a) Pure distilled water
    b) 5% Glucose solution (which is metabolized)
    c) Isotonic Saline (0.9% NaCl)
    d) Hypertonic Saline (3% NaCl)
    e) Half-normal Saline
42. **During a dental procedure, a patient experiences a "syncopal attack" (fainting). The rapid correction of blood pressure by the baroreceptors is an example of:**
    a) Positive feedback
    b) Feed-forward control
    c) Hormone regulation
    d) Volitional control
    e) Negative feedback
43. **A laborer working in 45°C heat collapses. His body temperature is 41°C, and he has stopped sweating. This "Heat Stroke" represents a failure of:**
    a) Positive feedback
    b) Active transport
    c) Negative feedback (Homeostasis)
    d) The mitochondria
    e) The glycocalyx
44. **A 5-year-old child presents with a "bloated" abdomen and thin limbs (Kwashiorkor). The abdominal swelling (Edema) is due to:**
    a) High plasma protein levels
    b) Decreased plasma colloid osmotic pressure
    c) Increased capillary hydrostatic pressure
    d) Blocked lymphatic vessels
    e) Increased salt intake
45. **A patient has a tumor in the Parathyroid gland (Hyperparathyroidism). Based on your reference, this will likely cause:**
    a) Hypocalcemia and Tetany
    b) Diabetes Mellitus
    c) Hypotension
    d) Muscle paralysis
    e) Hypercalcemia and bone resorption
46. **An infant is diagnosed with Cystic Fibrosis. This clinical condition is caused by a defect in the transport of which ion across the cell membrane?**
    a) Sodium
    b) Potassium
    c) Calcium
    d) Chloride
    e) Magnesium
47. **A person eats a very salty meal. Which homeostatic change will occur first?**
    a) Water moves from ICF to ECF (cells shrink)
    b) Water moves from ECF to ICF
    c) Blood pressure drops
    d) Urine output increases immediately
    e) Insulin levels rise
48. **A lab report shows a patient's plasma Potassium is 8 mEq/L (Normal: 4 mEq/L). The most dangerous clinical effect of this Hyperkalemia is:**
    a) Increased muscle strength
    b) Excessive sweating
    c) Cardiac arrest due to lack of repolarization
    d) Increased thirst
    e) Polyuria

49. **A patient with "Liddle's Syndrome" has overactive Sodium channels in the kidney. This would clinically present as:**
    a) Low blood pressure
    b) High blood pressure (Hypertension)
    c) Low blood glucose
    d) Weight loss
    e) Anemia
50. **In the "Intrinsic Pathway" of Apoptosis, the trigger is usually:**
    a) Binding of TNF to a surface receptor
    b) Extracellular bacteria
    c) Excessive water intake
    d) Muscle contraction
    e) Mitochondrial damage or DNA damage

Answer Key Table

Q#	Key	Q#	Key	Q#	Key	Q#	Key	Q#	Key
1	B	**11**	E	**21**	D	**31**	D	**41**	C
2	C	**12**	E	**22**	D	**32**	C	**42**	E
3	D	**13**	C	**23**	C	**33**	D	**43**	C
4	E	**14**	C	**24**	D	**34**	D	**44**	B
5	B	**15**	A	**25**	C	**35**	D	**45**	E
6	A	**16**	D	**26**	D	**36**	C	**46**	D
7	C	**17**	E	**27**	A	**37**	B	**47**	A
8	E	**18**	C	**28**	C	**38**	D	**48**	C
9	D	**19**	D	**29**	D	**30**	D	**49**	B

10	D	**20**	D	**30**	D	**40**	A	**50**	E

Explanations & Guyton References

Conceptual

- **Q21 (Osmosis Scenario):** In a hypertonic solution (350 mOsm/L vs 282 mOsm/L), water follows the osmotic gradient and leaves the cell. Guyton defines osmosis as the net diffusion of water caused by a concentration difference.
- **Q23 (Control System Gain):** The effectiveness of a system in correcting a stimulus is its "Gain". $Gain = Correction / Error$.
- **Q25 (Membrane Permeability):** At rest, "leak" channels allow K^+ to exit the cell much more easily than Na^+ enters. This high relative permeability to K^+ is why the resting potential is close to the K^+ Nernst potential (-94 mV).
- **Q31 (Ion Selectivity):** K^+ channels are specific because the carbonyl oxygens in the selectivity filter strip water from K^+ ions, allowing them to pass. Na^+ ions are too small to interact effectively with these oxygens and remain hydrated (and thus too large) to pass.
- **Q35 (Vmax):** Simple diffusion rate increases linearly with concentration. Facilitated diffusion involves a carrier protein with a limited number of binding sites; therefore, it reaches a maximal rate (V_{max}) when all sites are saturated.

Clinical Scenario Qs

- **Q41 (Isotonic Saline):** To expand ECF without affecting ICF, you need a solution with the same osmolarity as body fluids. 0.9% NaCl is isotonic and will stay in the ECF compartment.
- **Q42 (Negative Feedback):** The baroreceptor reflex is a classic homeostatic mechanism that opposes the initial stimulus (lowering BP if it's too high, or raising it if it drops) to restore stability.
- **Q44 (Edema/Kwashiorkor):** Plasma proteins (like Albumin) provide the "colloid osmotic pressure" that pulls fluid from tissues back into capillaries. In protein-malnourished children, this pressure drops, and fluid stays in the tissues (Edema).
- **Q47 (Salty Meal):** Increasing salt in the ECF creates a hypertonic environment, causing water to be drawn out of the cells via osmosis. This results in intracellular dehydration (cell shrinkage).
- **Q48 (Hyperkalemia):** Excessive ECF Potassium keeps cells in a depolarized state, preventing them from repolarizing to fire the next action potential. This is fatal in the heart, leading to cardiac arrest in diastole.